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Adaptive Cognitive Training with Reinforcement Learning

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Computer-assisted cognitive training can help patients affected by several illnesses alleviate their cognitive deficits, or healthy people improve their mental performance. In most computer-based systems, training sessions consist in graded exercises, which should ideally be able to gradually improve the trainee's cognitive functions. Indeed, adapting the difficulty of the exercises to how individuals perform in their execution is crucial to improve the effectiveness of cognitive training activities. In this paper, we propose the use of Reinforcement Learning (RL) to learn how to automatically adapt the difficulty of computerized exercises for cognitive training. In our approach, trainees' performance in performed exercises is used as a reward to learn a policy that changes over time the values of the parameters that determine exercise difficulty. We illustrate a method to be initially used to learn difficulty-variation policies tailored for specific categories of trainees, and then to refine these policies for single individuals. We present the results of two user studies that provide evidence for the effectiveness of our method: a first study, in which a student category policy obtained via RL was found to have better effects on the cognitive function than a standard baseline training that adopts a mechanism to vary the difficulty proposed by neuropsychologists; and a second study, demonstrating that adding an RL-based individual customization further improves the training process.

CCS Concepts: • **Computing methodologies** → **Reinforcement learning**; • **Applied computing** → **Health informatics**; • **Human-centered computing** → **User studies**; • **General and reference** → *Empirical studies.*

Additional Key Words and Phrases: Computerized Cognitive Training, Personalization, Adaptivity.

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1 INTRODUCTION

In the last two decades, several computerized cognitive training tools have been developed (see for example [1–6, 15, 27]). They can be effectively used for therapeutic purposes, for example for the rehabilitation of individuals that show signs of mild cognitive impairment [42][1]. A typical cognitive training program provided by these tools is composed by sessions prepared and led by a multidisciplinary team (including neuropsychologists, occupational therapists, and speech/language pathologists), which consist in exercises to practice in one or more cognitive domains, like memory, attention, language skills, or executive functions [31]. In addition to therapeutic purposes, cognitive

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[1]The term mild cognitive impairment denotes an intermediate stage between healthy aging and dementia that can be a serious consequence of several diseases.

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training can also be used to enhance mental performance in healthy people, for example to boost student learning [7], or enhance cognitive reserve of elderly people [24]. In this case, training sessions are not necessarily prepared by specialists and can become rather unstructured. Indeed, there are tools that allow the trainee to directly select the desired exercises among those made available.

Adapting the difficulty of the exercises to how individuals perform in their execution is crucial for the success of cognitive training programs [40, 41]. Therefore, existing systems for computerized cognitive training generally embed mechanisms for the execution of sessions consisting in graded exercises, whose difficulty increases gradually according to a set of predefined rules (generally defined by neuropsychologists), considering the results the trainees achieve. For example, several of such systems are able to automatically increment the difficulty when the trainee performance overcomes a predefined threshold, but the order in which the parameters characterizing the exercise are varied is fixed and predefined, and thus their relative importance for the single trainee is neglected. Even though they are often defined as adaptive, we believe there is an abuse of this term, as they do not take into account the specificities of the trainees.

In this paper, we give a contribution to the definition of adaptive mechanisms for cognitive training able to fully personalize the training path to the single individual. The general idea underlying our innovative proposal is embedding Reinforcement Learning (RL) agents [50] in computerized cognitive training systems, and associate a personalized agent to each trainee, for each cognitive exercise she/he performs. The agent interacts with the trainee while she/he is executing the exercise and, due to the peculiar characteristic of RL algorithms, learns from this experience a personalized policy able to vary the exercise difficulty according to how the trainee is performing. More in detail, the trainee's performance in the exercise is used as a reward to progressively elicit a policy able to modify over time the values of the parameters that determine the exercise difficulty, so that the trainee's performance is maximized in the long term. In this way, each exercise can be treated individually, and the policy used to vary its parameters in order to maximize the cognitive training effect can be learned independently for each trainee.

An optimal policy for a RL agent acting in a given environment is generally learned starting from a fully random policy, i.e., a policy assigning the same probability to all the possible actions the agent can execute. In our specific case, this means that all the possible exercise parameter variations would initially have the same chance to be executed, with two possible side-effects: (1) learning an optimal policy would need a long interaction between the trainee and the system, and would therefore be too time-consuming; (2) the initial randomness in the proposed exercises would cause a poor initial effect of the training process, resulting in the frustration of the trainee and probably in her/his demotivation in pursuing the cognitive training. In order to alleviate these inconveniences, we propose to associate *category policies* to exercises. A category policy should assign probability to actions that are not the true optimum for the single trainee, but that are closer to optimality for the category she/he belongs to with respect to a fully random policy. Starting from the exercise category policy, a shorter and smoother trainee-agent interaction would allow the refinement of optimal individual-tailored policies.

The first contribution of this article is a two-phase method for eliciting difficulty variation policies to be embedded in a cognitive training system, thereby achieving an adaptive experience for trainees. The purpose of the first phase is to build an exercise policy that is specific to a category of trainees, e.g., people over the age of 50 with no cognitive disorder, or college students under the age of 30, or patients suffering from mild cognitive impairment caused by multiple sclerosis. The second phase is dedicated to refining this policy and tailoring it to an individual in the category. We illustrate how to model the problem of learning a difficulty variation policy as a RL problem, how to construct a category policy associated to an exercise iteratively (i.e., by successive interactions with a dedicated RL agent of a set of homogeneous users who repeatedly execute the given exercise), and how to refine the category policy into an individual policy.

Our second contribution consists of two experiments that provide initial evidence for the effectiveness of the method we propose. In the first experiment, we show that a category policy, even though not specifically tailored to single subjects, can be effective enough to drive a successful cognitive training experience. Our demonstration is based on a study involving two groups of university students. The first group executed a training session composed by exercises whose difficulty-variation category policies were learned using our method. The second group executed the same session, but using a baseline difficulty-variation mechanism proposed by experts in cognitive training. The neuropsychological *PASAT* test [29, 52] was administered to the two groups before and after the intervention. While both groups significantly improved their scores in the test, the group trained using the student category policy performed significantly better than the other. In the following experiment, we applied the second phase of our method and refined individual policies for some students having the same characteristics of those previously involved. Although the students in this experiment did not improve significantly in the PASAT test, they scored the same as those trained only with the student category policy, but with significantly shorter training in one exercise, which demonstrates that RL-based individual personalization improves the training process compared to the exclusive use of a category policy. While our long-term goal is to run evaluations over multiple categories, in this paper we start with a first evaluation focused on one category, namely students, as a preliminary way to evaluate the validity of our method.

The rest of the paper is structured as follows. Section 2 gives an overview of the research connected to our work. Section 3 describes our method based on RL to build policies for exercise difficulty variation, illustrates its technical details, and presents the research hypotheses on which its effectiveness is based. In Section 4, we describe the two user studies we performed, and discuss how they support some of our hypotheses. Finally, Section 5 concludes the paper and highlights future work.

2 RELATED WORK

In the first part of this section, we focus on adaptability in computerized cognitive training systems, illustrating how some existing systems implement exercise difficulty variation mechanisms, and highlight their limits in providing a full adaptation to the trainee.

Then, we focus on two related research fields where adaptive training is fundamental and techniques based on Reinforcement Learning have been used. The first field is that of Intelligent Tutoring Systems (ITSs) [48], which exploit adaptive training and testing to facilitate the learning of a certain topic. Even if computerized cognitive training systems have some aspects in common with ITSs, they are intrinsically different. Indeed, cognitive training systems address basic cognitive abilities, like attention and memory, and do not directly exploit the testing of these skills in adapting the training process to the trainee. Testing is administered separately (before and after the training) in order to assess improvements, by means of validated test batteries. The *PASAT* test [29, 52] we used in our experiments is an example of a validated battery which can be used to assess a subject's information processing speed, her/his working memory, and her/his sustained attention. The second research field is that of interactive systems for physical training and motor rehabilitation. These tools are similar to cognitive training systems because even in this case training and rehabilitation should adapt to individual movements and reactions.

In addition to the research areas mentioned above, which are the closest to our research, there are several other healthcare applications where RL can be effectively exploited to improve adaptability. The interested reader can refer to two recent reviews [21, 58] for more information and other applications of RL in healthcare systems. Here we just mention a study [44], focusing on cognitive deficits, that illustrates a system integrating temporal constraint reasoning with RL to support people in reminding them their daily activities. Differently to ours, the system presented in [44] is an assistive/compensative tool, while we aim at restoring the impaired cognitive functionality with a personalized intervention.

2.1 Exercise difficulty variation in computerized cognitive training systems

Many systems for computerized cognitive training offer mechanisms to adapt the difficulty of exercises to the trainee performance in their execution. To give a comprehensive view of the state of the art, we analysed several systems designed for rehabilitation of mild cognitive dysfunctions. We mainly focus on tools for health professionals, but also consider some experimental platforms dedicated to researchers.

Brainer [1] is a rehabilitation software that can be used to treat several diseases. It has been used in various hospitals and clinics, especially with patients suffering from schizophrenia [17]. The clinicians can manually assign to patients exercises of three predefined levels of increasing difficulty; at each level, the patient has on average five attempts to pass to the next level. The rehabilitation can be conducted at the hospital, under the supervision of a specialist, but also at home, where the patients have to be followed by a caregiver. This may strongly limit the patient's autonomy in the rehabilitation.

NeuroTracker [5] is a cognitive training program designed to improve mental performance in various contexts, such as sport [33], or old age [9]. It provides a single 3D visual exercise, where users see a number of yellow balls moving around their screen, and are asked to track the highlighted balls. The main parameter influencing the difficulty of the exercise is the ball speed, which is adjusted closely to each individual's tracking ability using an adaptive staircase algorithm [53]. Every time the user is able to correctly track all the highlighted balls, the ball speed increases, while, when she/he does not capture all the targets, the speed decreases. The speed fluctuation gets smaller as the session goes on, thus adapting to each user's skill level.

RehaCom [6] is an advanced cognitive training software usable for several diseases on a common personal computer, with a normal or a special keyboard, dedicated to patients with serious physical or cognitive impairments. It has been used to treat auditory hallucinations [11], to improve cognitive abilities of schizophrenic patients [23, 34], to train children with attention-deficit hyperactivity disorder [8], and to implement a remediation therapy in acute depressive episodes [46]. It offers 25 different training exercises (called modules) grouped in cognitive domains. The exercise difficulty is modulated through various parameters. The initial difficulty level for the exercises is selected by the therapist, according to the patient's characteristics and deficits. As the training progresses, the tasks become easier or harder depending on the patient's performance, using predefined rules that are not dynamically adjustable. At the end of the training, the software offers results on, for example, the achieved progression level, the number of mistakes made, or the time taken for each task. By analyzing the data, the therapist is able to further address the training.

CogniPlus [3] is a tool developed for the treatment of cognitive function disorders. It has been used to treat attention deficits in children with hyperactivity disorder [54], in conjunction with physical training to support elderly people in activities of daily living [30], and to train cognitive functions in patients with Parkinson's disease [55, 61]. It offers 16 types of exercises, each of which has numerous levels of difficulty, covering all areas of cognitive training. The exercise difficulty depends on the values of the parameters characterizing the exercise, such as the semantic similarity of stimuli, or the maximum reaction time to them. The initial difficulty of some exercises can be automatically selected by the system, by measuring the performance of the patients in the starting exercise. The progression from one level of difficulty to the next is based on achieving a minimum performance in the exercises, and follows a fixed sequence of grades.

The HappyNeuron Pro platform [2] includes tools used for the treatment of various cognitive disorders including depression [14, 35], and Huntington's disease [56]. The platform offers 38 different training exercises across 9 areas of cognition. Exercise parameters within each exercise can be customized to tailor cognitive practice at a difficulty level that suits each individual's cognitive and learning needs. The level of exercise difficulty automatically progresses after the user achieves 100% accuracy on two consecutive trials. The automatic progression of exercise difficulty follows predefined patterns but the exercise parameters, which influence the difficulty, can also be manually fine-tuned by the medical staff, for example when treating serious mental illnesses, where the step from one level of difficulty to the next may be excessive.

Erica [4] is a neuropsychological platform for cognitive rehabilitation. The provided exercises are dedicated to the rehabilitation of specific skills such as attention, spatial cognition, memory, and verbal/non-verbal executive functions. The platform is suitable for patients suffering from neuropsychological deficits resulting from degenerative diseases, psychiatric disorders, brain injury, and evolutionary disorders. Erica allows the rehabilitator to adapt the parameters of each exercise (e.g., type of stimuli, how they are presented on the screen for how long, or presence of distractors) to the patient's profile. The rehabilitator is supported by the results of the previous sessions in monitoring the patient's progress and properly escalating the difficulty in case of improvement. However, all the exercise sessions must be set manually by the medical staff, as the system does not support any automated variation of exercise difficulty.

The Padua Rehabilitation Tool (PRT) [15] is an experimental platform usable by patients regardless their pathology. For example, it has been used to treat dementia [16]. PRT comprises 35 exercises for the cognitive stimulation of various functions: attention, memory, vocabulary, logical reasoning, recognition, orientation, and motor control. Each exercise is structured in levels of increasing difficulty. The main variables that vary are the number of stimuli, the time they are shown to the trainee, and their colors or shape. Only one variable at a time is modified to increase the difficulty of the exercise. In case of a correct response, the exercise automatically goes to the next level by applying fixed rules. In case of an error, the patients can retry the same level up to three times.

MS-rehab [27] is an experimental system designed for the cognitive rehabilitation of patients suffering from multiple sclerosis. It provides exercises, designed by a team of computer scientists and neuropsychologists, that cover the three main cognitive domains: attention, memory, and executive functions. Each exercise has multiple versions in which the stimuli vary, for a total of 52 exercises. As for many of the previously described systems, the exercise difficulty depends on the values of the parameters (e.g., number of stimuli, their color, maximum completion time, presence/absence of simple/complex distractors) characterizing the exercise itself. The clinicians can live-monitor the exercise execution and possibly vary its difficulty according to the patient performance. MS-rehab also embeds an automatic mechanism for difficulty variation defined by expert rehabilitators: difficulty is increased whenever the patient performance on an exercise overcomes the 80% of the maximum for two consecutive times.

2.1.1 Difficulty adaptation and personalization. Table 1 shows a comparison of the above-described systems, according to two main dimensions: the availability of manual and/or automatic mechanisms to select a tailored initial difficulty for a cognitive training exercise, and whether the difficulty of the next exercise instance can be manually or automatically varied. As for the automatic variation, we distinguish between *fixed-grades* variation, and *adaptive* variation. By fixed-grades variation, we mean that the system is able to vary the difficulty according to how the trainee performs, using a fixed sequence of levels based on pre-defined rules. By adaptive variation, we mean that the way the system changes the exercise difficulty can be different from trainee to trainee. To make an example of fixed-grades versus adaptive variation, imagine a fictitious exercise for attention training where the user has to click only on a target image among those sliding on the screen, and that the exercise difficulty depends on two parameters: the total number of images shown to the user and their sliding speed. We say that a system providing such an exercise is fixed-grades if, for example, it increases the number of images and their sliding speed of fixed amounts, whenever the user executes an exercise instance without errors. This implies that any user passing the exercise will get exactly the same next exercise. In other words, the systems adapts to the user performance, but the way it adapts is the same for all the users. On the contrary, a fully adaptive difficulty variation mechanism is able to vary the difficult according to which user interacts with the system. Coming back to the previous example, *user A* could better improve her/his cognitive performance if the system first increased several times the number of images, and later started augmenting the sliding speed, while for *user B* it could be the other way around. A truly adaptive system would be able to use the first difficulty variation pattern for *user A*, and the second for *user B*.

System	*Initial difficulty for trainee profile*		*Difficulty variation*		
	Manual	*Automatic*	*Manual*	*Automatic*	
				Fixed-grades	*Adaptive*
Brainer	✔	✘	✔	✘	✘
NeuroTracker	✘	✘	✘	✘	✔
RehaCom	✔	✘	✘	✔	✘
CogniPlus	✔	✔	✘	✔	✘
Happy Neuron Pro	✔	✘	✔	✔	✘
Erica	✔	✘	✔	✘	✘
Padua Rehabilitation Tool	✔	✘	✘	✔	✘
MS-rehab	✔	✘	✔	✔	✘

Table 1. Difficulty setting in computer-based cognitive training systems (✔ supported; ✘ not supported).

A common characteristic of almost all the systems we analyzed is that the training path for a subject can start at a difficulty level that is compatible with her/his cognitive profile. In other words, the therapist can usually select the initial set of exercises by selecting those that are congruous to the cognitive domains the subject needs to strengthen, and personalize the exercises according to her/his condition and/or disease. NeuroTracker and CogniPlus are two exceptions: in CogniPlus the initial difficulty of some exercises can also be automatically determined by measuring the performance of the trainee in an initial testing exercise, while in NeuroTracker the initial difficulty is the same for all the trainees. Nevertheless, NeuroTracker is the only system that provides adaptive difficulty variation, thanks to the adoption of the staircase algorithm, which adjusts the speed of balls moving on a 3D space according to the performance of the person who is attempting to track them. A limit of the NeuroTracker approach is that it cannot be easily extended to more complex exercises where the difficulty level depends on several factors and not only on speed, for example exercises in which the trainee is presented multiple stimuli possibly having several values each.

None of the other systems presented above is truly adaptive, according to the definition we gave. Indeed, in these systems the progression of the exercise difficulty is decided through the intervention of clinicians, or through the application of automatic rules that are triggered when the trainee overcomes a certain performance threshold. These rules are fixed and generally based on predefined patterns. For example, the system MS-rehab increases the difficulty by modifying the exercise parameters in a predefined order or with predefined probabilities, and each parameter according to a predefined sequence of values. Therefore, the progression of the exercise does not take place based on the actual ability of the trainee with respect to the individual parameters, and strongly limits the achievable progressions.

Finally, basing the progression of the exercises on fixed and predefined rules, all these softwares only partially allow the trainee to work at a level that has the maximum training effect since these rules, even though usually defined with the contribution of domain experts, are non personalized and therefore they can be suboptimal for single individuals.

2.2 Adaptability in intelligent tutoring systems

Intelligent Tutoring Systems (ITSs) [48] are educational software systems that contain an artificial intelligence component. Like computerized cognitive training systems, ITSs exploit intelligent interfaces to adapt their behaviour to different subjects, nevertheless their goals are intrinsically different. While ITSs facilitate the learning of a particular educational field (e.g., mathematics [28], computer science [20], or language [18]) through adaptive training and testing, cognitive training systems try to optimize basic cognitive abilities, like attention and memory, exploiting cognitive plasticity. In other words, the latter are not aimed at learning a specific subject, but at improving the basic functionality of the brain that is needed to perform all our activities, including learning. It is worth noting that cognitive training does not include testing, and the effects of training sessions are measured through the administration of specific and validated test batteries. These batteries are generally cognitive exercises that must be different from those used for training. In fact, if subjects were tested on the same exercises used for training, the only evidence that would result would be that they were able to solve the proposed exercise, not that their cognitive abilities were improved. On the other hand, in ITSs the test of the student's knowledge of the specific subject being taught is often exploited as feedback to the system to automatically improve its tutoring ability. However, since both ITSs and computer-based cognitive training share the goal of adapting the learning/training process to a specific user, and ITSs and our cognitive training solution share the use of artificial intelligence algorithms as the driver of their adaptability, it is interesting to illustrate how AI-based adaptability has been implemented in some ITSs and how these approaches compare to ours.

In a typical ITS architecture [39], the component that adapts the learning material to the specificities of the student, by interacting with the *student model* and the *expert knowledge module*, is called the *tutoring module*. Various artificial intelligence techniques have been used in literature for the implementation of this component. We refer the reader to the several comprehensive reviews on ITSs for further details (see for example [36], or [22]), and we restrict here the discussion on some recent works where the *tutoring module* is based on RL, which is the same technique we adopt in our approach to cognitive training, or on a combination of RL with other AI techniques.

In [26], the idea is that tutoring rules can be, firstly, learned off-line with Artificial Neural Networks (ANNs) by observing a training set of human tutors' behaviors and, then, adapted at run-time with RL, by observing how each learner reacts at different states of the learning process. The approach was tested with 5-6 year old children, who used an educational game running on a tablet. The RL algorithm adopted in this research is SARSA [50] (the same we used for our research in this paper) and was used to adapt the difficulty of the game, in a quite similar way to how the difficulty of our cognitive training exercised is modified. Differently to us, in [26], an initial tutoring model is learned via ANNs and then personalized with RL, while we adopt RL also for building the initial model.

In [59], the authors proposed and applied a Hierarchical Reinforcement Learning (HRL) framework to induce a hierarchical pedagogical policy, which makes decisions first at the level of the problem proposed to the student, and then in the selection of the next problem step. Results from an empirical classroom study showed that the HRL policy was significantly more effective than an algorithm that combines deep neural networks with RL, and than a random baseline step-level policy. Therefore, the results suggest that HRL can be more effective than flat RL in pedagogical policy induction. In our research, we focus only on the step-level, trying to optimize the next configuration of a pre-selected cognitive training exercise. However, using HRL could be an option for us to implement automatic configuration mechanisms for selecting appropriate exercises to be inserted in the training sessions, instead of leaving the choice to the trainee or to an expert.

Another interesting approach to improve adaptability in the field of ITSs is described in [47]. This research investigated the impact of immediate and delayed reward functions on RL-induced policies, and evaluates the effectiveness of such policies within an ITS called Deep Thought. Interesting to us, an experiment was done dividing students into *fast* and *slow* learners depending on their average response time on the initial tutorial level. The research found that there was a significant interaction effect between induced policies and students' incoming proficiency. More specifically, fast students could learn equally well regardless of the policy employed by the RL-based tutor. On the other hand, slow students learned as much as their faster peers if effective strategies were used, while they learned significantly less if coached with ineffective strategies. This result supports our idea of diversifying cognitive training policies according to trainee categories and suggests that adaptation is particularly relevant for "weak" subjects, for example subjects suffering from cognitive deficits caused by a disease or old age.

2.3 Adaptability in physical training and motor rehabilitation systems

Similarly to cognitive training, adaptation and personalization to the subject are essential aspects of physical training and motor rehabilitation. The general hypothesis of research on these two research fields is similar to ours: adaptation driven by artificial intelligence (in particular RL) is able to improve a specific skill of people (i.e., physical activity proficiency vs cognitive performance) more than a static or expert based policy. Therefore, we highlight some specific researches where RL has been successfully applied to improve adaptability, this time in physical training and motor rehabilitation systems.

Some mobile apps [57, 60] help patients increase their level of physical activity using personalized suggestions produced by RL algorithms. In more details, in the study presented in [57], some sedentary diabetes type 2 patients were provided with a smartphone-based pedometer and a personal activity plan. These patients were sent short messages to encourage activity, with a frequency between once a day and once per week. The sending of these message were personalized through a RL algorithm, in order to improve each participant's compliance. The algorithm was compared with a static policy for sending messages. People who received messages generated by the RL algorithm increased the amount of activity and pace of walking, whereas the patients in the control group did not. A similar research is presented in [60], where a fitness app called `CalFit` uses a RL algorithm to generate goals, both personalized and achievable, related to the number of daily steps to be taken. A user study demonstrated that a group of university students using `CalFit` to receive personalized step goals increased the daily step count after a 10-week intervention, while the control group (receiving goals of 10,000 steps/day) had a significant decrease.

A challenging area of research in which a strong degree of adaptability is needed is neurological rehabilitation, which aims to restore control of voluntary movements, to improve patients' ability to be independent in activities of daily living, and to prevent further disability [37]. A typical example is gait training/rehabilitation [25, 32], where RL is used to train healthy subjects to adopt a more natural and efficient gait pattern. The papers demonstrates that, if the gait rehabilitation is grounded on RL, the training program is more effective.

In [10], a RL-based adaptive algorithm is used to enhance existing virtual reality-based motor rehabilitation platforms by tailoring the response of the games to the patient changing needs.

In particular, the patient's control and speed are dynamically interrogated to adjust the game difficulty, in a way that is similar to how we use the trainee's performance as a reward to adjust the difficulty-variation policy of our cognitive training system. Other examples of RL-enabled motor rehabilitation are presented in [38, 43].

Brain-machine interfaces are another tool that can be used in motor rehabilitation, when physiological and pathological aspects limit brain communication channels. Adaptability is a crucial issues for this type of solutions, a simulation presented in [13] shows how RL can be used for adaptive threshold control of a restorative brain-computer interface. An interesting recent survey [12] presents how machine learning techniques can be exploited in this context.

3 A REINFORCEMENT LEARNING MODEL FOR COGNITIVE TRAINING

The nature of cognitive training requires continuous consideration of the feedback the trainee provides in order to develop truly adaptive cognitive training systems. Indeed, test batteries can be used to evaluate attention or memory levels, but the reaction of individuals to different stimuli remains unknown and subjective. In this context, Reinforcement Learning seems to be a natural solution, being the learning process based on reward loops. In our proposal, adaptation is achieved by exploiting the learning-from-experience ability of RL algorithms, which we use to learn a policy able to vary the exercise difficulty according to how the trainee is performing, in order to maximize the long-term exercise training effect. Following this approach, the learned policy can be different from trainee to trainee, since we consider how each parameter influencing the exercise difficulty impacts on the subject performance in the learning process. Therefore, starting from a training program (possibly) chosen according to the cognitive profile of the trainee, her/his training experience can be driven by an exercise progression where the exercise difficulty is changed depending on how the single subject is performing over time.

This degree of adaptability for individuals can be achieved by introducing a RL agent for each exercise–trainee pair. Another type of adaptability is associated to categories of trainees, e.g., people over 50 without any cognitive disorder, or under 30 university students, or patients suffering from mild cognitive impairment caused by multiple sclerosis, or children with the same special need. The necessity of category personalization arises from the fact that each cognitive disorder has its specific way to inhibit distinct cognitive functions, or simply that cognitive abilities are shaped differently according to demographic and/or social characteristics. Exercises can be adapted to categories of individuals introducing a RL agent for each exercise–category pair. In both cases, the RL problem can be modeled in the same way, as described below.

3.1 RL problem description

Our RL problem was to create agents able to learn policies that automatically regulate the difficulty of their associated cognitive training exercises, so that the trainee's performance in their execution is maximized. In this section, we informally describe the problem, while in Section 3.2, we illustrate its formalization in details.

The diagram in Figure 1 summarizes the interaction between the components of our RL problem. The state of the environment consists in the values of the parameters that determine the difficulty of an exercise. With this information, the computerized cognitive training system MS-rehab [27] is able to generate a new training exercise. Figure 2 shows various phases of an example exercise (among those provided by MS-rehab), aimed at training alternating attention. In the exercise, the trainee is shown two images (the goat and the sparrow in this example) and is asked to select the occurrences of the first (the goat) among the images sliding on the screen from right to left, until a sound is played. At this point, the trainee has to change the target and select the second image (the sparrow). When the sound is played again, the trainee has to change target again, and so on for a

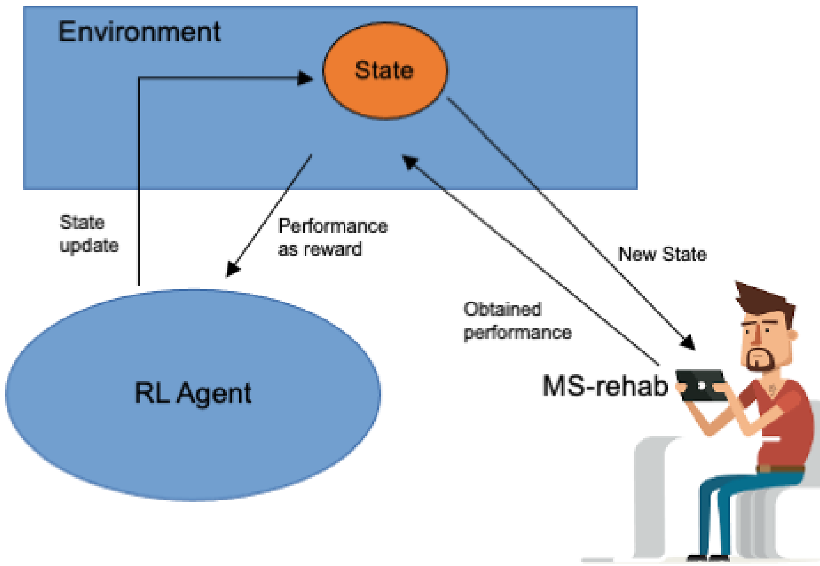

Fig. 1. Interaction between the components of our RL problem. The user trains with the computerised cognitive training system MS-rehab and is provided with exercises tailored to his/her performance.

certain number of iterations. The top left box of Figure 2 shows a missed target: the trainee was not able to click on the goat, and the image was highlighted in yellow. The top right box shows instead a correctly captured target (green highlight). The bottom left box shows an error (red highlight), while the bottom right box visualizes a correct answer after the sound has been played. Finally, the last box illustrates the feedback the system gives to the trainee, showing the performance obtained in the exercise.

After the trainee has completed an exercise execution, her/his obtained performance (i.e., how good she/he was in executing the exercise) is communicated back to the environment that, in turn, converts it to a reward, which is transmitted to the RL agent. The agent takes care of modifying the value of the parameters according to the received reward. The agent can decide to increase a parameter value, to decrease it, or possibly decide that the trainee needs to perform another exercise with the same difficulty. With reference to the exercise in Figure 2, the agent, for example, can increase (or decrease) the time interval between two sliding images, or change the number of iterations, or modify another parameter. The action performed by the agent modifies the state of the environment and, as a direct consequence, instructs MS-rehab to build another exercise. Then, the trainee executes the new exercise and the agent-environment-MS-rehab loop iterates. The loop can potentially continue forever or, more realistically, until the trainee has reached a high skill in the exercise execution.

To give an example, suppose a trainee has executed an instance of the exercise in Figure 2. If the obtained performance is excellent, then the agent (based on the policy previously learned through the interaction with the user) may decide to set a parameter, in the next exercise instance, to a value that previously resulted particularly hard for the trainee. Alternatively, if the performance was good, the agent could instead decide to make harder the value of a parameter that previously had not given problems to the trainee. Conversely, should the performance be poor, the agent would

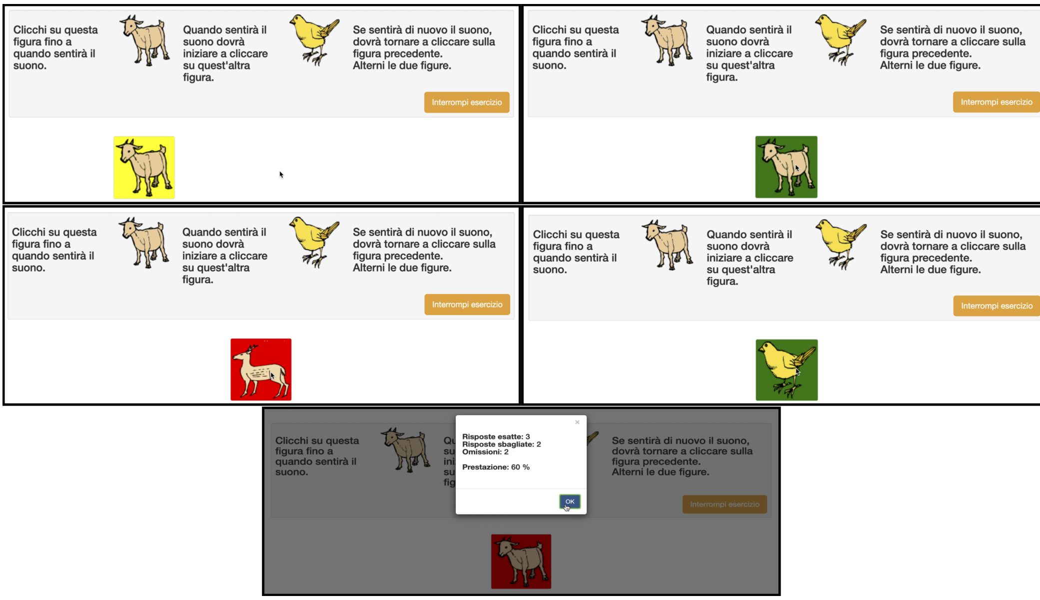

Fig. 2. Exercise for alternating attention training (*Clicchi su questa figura fino a quanto sentirà il suono* = Click on this image until you hear the sound; *Quanto sentirà il suono, dovrà iniziare a cliccare su quest'altra figura* = When you hear the sound, you will have to start clicking on this other image; *Se sentirà di nuovo il suono, dovrà tornare a cliccare sulla figura precedente. Alterni le due figure* = If you hear the sound again, you will have to go back to clicking on the previous image. Alternate the two images; *Interrompi esercizio* = Interrupt the exercise; *Risposte esatte* = Correct answers; *Risposte sbagliate* = Wrong answers; *Omissioni* = Omissions; *Prestazione* = Performance).

probably decide not to increase the exercise difficulty but to decrease it, in particular if the trainee has obtained insufficient results several times.

3.2 RL problem formalization

In this section, we formally describe the elements of the RL problem introduced in Section 3.1. For the sake of clarity, we refer to the specific exercise for alternating attention presented in Section 3.1, but the problem can be modeled similarly for other training exercises.

State. The difficulty of the exercise varies according to five parameters, which can have three different values each, corresponding to three levels (*easy*, *medium*, and *difficult*) of increasing difficulty.

- **#iterations**: the number of times (1, 2, or 3) the trainee has to change the target between the two images;
- **#stimuli**: the number of images (20, 15, or 10) shown in an iteration;
- **time**: the time (4, 3, or 2 seconds) an image remains on the screen before being replaced by the next one;
- **distractors**: the presence of distracting animations in the background (distractors can be *absent*, *simple*, or *complex*);
- **color**: the color (*full*, *faded*, or *black and white*) of the images.

The state of the environment is composed by a 5-ple of pairs (**parameter**, *value*). For example:

$$\langle(\textbf{\#iterations}, 1), (\textbf{\#stimuli}, 15), (\textbf{time}, 3),$$
$$(\textbf{distractors}, \textit{absent}), (\textbf{color}, \textit{bw})\rangle$$

In this state, the parameters **#iterations**, and **distractors** are at the *easy* level, **#stimuli** and **time** are at the *medium* level, and **color** is at the *difficult* level.

Actions. The RL-agent can execute actions that change at most the value of a single exercise parameter. This choice was made mainly to allow the use of RL algorithms without approximation functions, and therefore to model the problem in such a way that an exact policy can be elicited. Limiting to one the number of parameters that can be modified at a time has strongly limited the maximum number of possible action-state pairs, thus allowing an implementation that does not require an approximation function. For the example exercise given above, possible actions can be "decrease **time** to 2 seconds" , or "insert a *simple* **distractor**", or "increase **#iterations** to 2", or "don't change any parameter". In total, for that exercise, there are 11 possible actions: one to increase each parameter, one to decrease each parameter, and the action that leaves all parameters unchanged.

Policy. The RL-agent policy is represented as a $\#actions \times \#states$ matrix π, where each entry $\pi_{a,s}$ is the probability of executing action a in state s. The probability $\pi_{a,s}$ is proportional to the *value* the agent will get in the future if it executes action a in state s. For example, in the state above, action "decrease **time** to 2 seconds" could have the highest probability, action "insert a *simple* **distractor**" could have the second highest probability, action "don't change any parameter" could have a small, not null, probability, and all other actions could have probability equal to 0. When the agent has completed the learning, it exploits the final policy, and varies the difficulty of the next exercise instance by selecting the action having the maximum probability. In our example, the agent would decrease to 2 seconds the time an image remains on the screen. When learning, the agent uses the reward from the environment to refine its policy, in a way determined by the adopted RL algorithm (see below for details): when the agent receives the reward, it modifies the probability $\pi_{last\_action,last\_state}$ in the policy matrix π, increasing it if the reward is high, decreasing it otherwise. This means that the value of action $last\_action$ in state $last\_state$ increases if the execution of $last\_action$ was very rewarding, and decreases if it brought a low reward. We take into account the difference between long term and short term reward using a discounting mechanism: the reward given to the agent is scaled proportionally to the deepness in the action tree. To exemplify, the value of the first action selected by the agent is more important than the actions taken afterwards.

Reward. The reward obtained by the agent in response to an action is the performance of the trainee in the exercise obtained by executing the action. The performance calculation takes into consideration 3 variables:

- c: number of correct answers given by the trainee;
- w: number of wrong answers given by the trainee;
- m: number of answers missed (not given) by the trainee.

To define the formula for the performance, we introduce *precision* ($p = \frac{c}{c+w} \in [0, 1]$) and *recall* ($r = \frac{c}{c+m} \in [0, 1]$). The performance *per* of a trainee is nothing but an *F-measure*:

$$per = (1 + \beta^2) * \frac{p * r}{\beta^2 * p + r} \in [0, 1] \quad (1)$$

The parameter $\beta > 0$ weights precision and recall in the calculation of the performance. We set $\beta = 1$, a typical value which gives equal importance to precision and recall. For example, suppose that in the execution of an instance of our example exercise, the trainee has correctly captured 15 images, has wrongly selected 4 images, and has missed 5 images she/he should have captured. Then, her/his performance can be calculated by equation 1 using $p = \frac{15}{15+4} = 0.79$, and $r = \frac{15}{15+5} = 0.75$, thus obtaining $per = 0.77$.

RL algorithm. The algorithm used by the agent to learn the exercise policy is based on tabular *SARSA* [50]. *SARSA* is an on-policy method (i.e., it attempts to improve the policy π that is used to make decisions about the action to take) that learns an action-value function $Q_\pi(s, a)$ for the current policy π of the agent, and for all states s and actions a. $Q_\pi(s, a)$ refers to the long-term return of the current state s, taking action a under policy π. In each state, the agent choose a new action in a probabilistic way with an $\epsilon - greedy$ evaluation of Q, where ϵ (called the exploration parameter) is the probability of choosing an action different from that with the highest value of Q. The value update has the form:

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha \left[R_{t+1} + \lambda Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t)\right]$$

where $Q(S_t, A_t)$ is the value of Q at time t for the state-action pair (S_t, A_t), α $(0 \leq \alpha \leq 1)$ is the step-size parameter and λ is the discount factor. The update is done after every transition from a non-terminal state S_t to S_{t+1}, and if the latter is terminal, then $Q(S_{t+1}, A_{t+1}) = 0$. The update formula uses every element of the quintuple $(S_t, A_t, R_{t+1}, S_{t+1}, A_{t+1})$ in making up a transition from one state-action pair to the next and led to the name *SARSA*.

SARSA was chosen as the learning algorithm mainly because the problem could not be effectively divided into easily recognizable separate episodes, and so it was decided to learn at each step of the interaction with the environment (i.e., after completing every single exercise). Furthermore, the tabular version was chosen by virtue of the limited number of possible states. In the most complex exercises provided by MS-rehab, in fact, there are six parameters, and each of them can vary between three different values, for a total of 3^6 possible states. This number is feasible for a solution without approximations, and consequently the same applies for exercises with a lower number of parameters. In order to assure a sufficient exploration of the state-action space, it is important to favour exploration by setting the exploration parameter ϵ of *SARSA* to a sufficiently high value. A way to check if the agent has explored enough states is to monitor the value of the overall difficulty level of the exercise and verify it has reached the maximum.

3.3 Learning adaptive policies

We propose a method to elicit policies to be embedded in a cognitive training system to obtain an adaptive experience for trainees. The method includes two phases: the purpose of the first phase is to build a policy for an exercise that is specific for a category of trainees; the second phase is dedicated to refine the policy elicited in phase 1 and tailor it to a single individual of the category. The first phase of our method is composed by the following steps:

(1) A number of subjects in one category are selected. It is important to inform these subjects that the purpose of their interaction with the training system is not to improve their personal cognitive capacities but, instead, to instruct the system so that it becomes able to profitably train the cognitive abilities of future users;
(2) The first subject executes a session including several instances of the exercise, starting from the lowest difficulty level; the exercise–category agent starts with a fully random policy[2], and adjusts the policy, using the *SARSA* algorithm, according to the interaction with the subject;

[2] With the term *fully random policy* we denote a policy having the same probability for all action-state pairs in matrix π.

(3) Successively, the exercise–category agent interacts in the same way with the other subjects starting from the policy learned by interacting with the previous one, thus it progressively refines the policy;
(4) The policy obtained after all the subjects have executed their exercise sessions is selected as the exercise–category policy.

The second phase of our method involves personalizing and adapting the exercise–category policy to an individual who needs to improve her/his cognitive abilities. The subject interacts with the exercise–category agent, which further adapts the policy by exploiting again the *SARSA* algorithm.

The rationale behind our method is that learning an effective individually personalized policy from an initial fully random one would imply an initial random selection of the actions to be performed by the RL agent, which would result into a randomness in how the exercise difficulty is changed, and therefore in a non-optimized experience for the trainee. Even if it is true that the policy would adapt overtime, the initial stage of the training process would probably be ruined by such a random selection of the exercise difficulty. Starting instead from a policy that is already adaptive for the category of users to which the trainee belongs, the interaction between the RL-agent and the trainee, even if not truly optimal for the single trainee, would proceed more smoothly, and permit to rapidly converge to an optimal individual policy.

The effectiveness of the method we propose is based on four hypotheses:

(1) A category policy learned using RL is able to drive the cognitive training of subjects belonging to that category better than the mechanisms currently adopted in available computerized cognitive training systems;
(2) An individually personalized policy learned for a subject belonging to a category would better drive the cognitive training for that subject than the category policy from which the individual policy derives;
(3) A policy elicited for a category would not improve the training process of subjects belonging to another category, if applied for their cognitive training;
(4) An individually personalized policy for a subject belonging to a category can be more efficiently learned refining a policy previously learned for her/his category instead of starting from a fully random policy.

In the next section, we investigate hypotheses 1 and 2, while the verification of the latter two hypotheses is left as future work.

4 EXPERIMENTAL EVALUATION

4.1 Hypothesis 1

Our first experiment aimed at evaluating an exercise–category policy built following the first phase of the method described in the previous section. The experiment involved some university students, and was designed in collaboration with researchers of the Department of General Psychology of the University of Padua.

4.1.1 Experiment design. This experiment and the one described in Section 4.2 involved subjects selected via convenience sampling who were not given any rewards. The experiment was structured as follows:

(1) Two cognitive training exercises were selected among those provided by the MS-rehab system: one for alternating attention, and the other for working memory; these exercises were chosen because they are challenging for young healthy subjects like the students involved in the experiment;

(2) A group T of 10 students trained 2 exercise–category policies by executing 20 instances of each exercise[3] using MS-rehab, starting from the lowest level of difficulty;
(3) Two other groups of students (group A and group B) were created; both groups included 10 subjects, different from those who trained the exercise–category policies;
(4) The *PASAT* neuropsychological test [29] was administered to all the subjects of groups A and B to measure their cognitive functioning before starting the training;
(5) All the subjects in groups A and B had sessions[4] of the two selected exercises using MS-rehab; group A used the version of MS-rehab embedding the difficulty-variation policies learned in step 2, while group B used the baseline version of MS-rehab without RL, where the difficulty variation mechanism was designed by experts in cognitive rehabilitation;
(6) The *PASAT* test was administered again to the subjects in the two groups in order to highlight any improvement compared to the pre-training administration, and any difference between the two groups;
(7) In addition to the results of the *PASAT* test, data were collected about the number of failed exercises, the number of committed errors, and the number of omitted answers by both groups, trying again to highlight any differences between them;
(8) Finally, after a week, all subjects in groups A and B were asked to perform a single instance of both exercises at the maximum difficulty level; the purpose of this further evaluation was to verify if one of the two groups showed a higher and lasting level of performance over time.

The experts' policy for varying difficulty provides that the difficulty is automatically increased whenever the student's performance exceeds 80% of the maximum for two consecutive exercises. This is a general approach also adopted in other systems, such as the HappyNeuron Pro platform [2], where the difficulty level of the exercise automatically progresses after the user achieves 100% accuracy on two consecutive trials. Successfully repeating an exercise twice is considered a sufficient condition to prevent the score from being due to randomness. On the other hand, in our case the experts decided to require a high but not perfect level of performance to indicate that the trainee is progressing solidly to the next level, but without overly penalizing few errors in the exercise. In fact, the way we calculate the subject's performance (see Section 3.2) takes into consideration any single error (e.g., the selection of a wrong target in the attention exercise) or omission (e.g, a delayed reaction to a stimulus in the memory exercise).

The chosen exercise for alternating attention is described in Figure 2, while the one for working memory in Figure 3. In this last exercise, the trainee looks at the images sliding on the screen. If the current image is the same as the one seen $N \geq 1$ images before, the trainee has to click on the green button, otherwise on the red button. The top left box of Figure 3 shows a correctly captured target (green highlight). The top right box shows an error (red highlight). The bottom left box visualizes a missed target (yellow highlight). Finally, the bottom right box illustrates the feedback the system gives to the trainee, showing the performance she/he obtained in the exercise. The difficulty of this exercise for working memory varies according to four parameters:

- **N**: number of images (1, 2, or 3) that the trainee must remember, having to compare the image currently on the screen with the one that appeared N images before;
- **#stimuli**: the total number of images (20, 40, or 80) that are shown in an exercise;
- **time**: the time (5, 4, or 3 seconds) an image remains on the screen before being replaced by the next one;
- **color**: the color (*full*, *faded*, or *black and white*) of the images.

[3]This number has been selected to allow users to potentially reach the maximum level of difficulty of the exercise.
[4]By *session*, we mean the execution of a series of instances of an exercise, starting from the lowest difficulty up to the highest level.

Fig. 3. Exercise for working memory training (*Osserva le figure che scorrono sullo schermo* = Observe the images sliding on the screen; *Se le figura corrente è uguale a quella vista 1 figura prima, allora clicca il tasto verde, altrimenti clicca il tasto rosso* = If the current image is the same as the one seen 1 image before, then click the green button, otherwise click the red button; *Interrompi esercizio* = Interrupt exercise; *Risposte esatte* = Correct answers; *Risposte sbagliate* = Wrong answers; *Omissioni* = Omissions; *Prestazione* = Performance); *Ottimo! Esercizio superato!* = Excellent! Exercise passed!

The demography of groups T, A, and B is shown in Table 2. In particular, there are entries related to age, gender, school years, and ability with video games (especially those running on web platforms). This last entry was included because the training exercises provided by MS-rehab resemble video games, and it is based on a scale of integers between 0 and 3: 0 means *video games never or almost never used*, 1 means *video games occasionally used*, 2 means *video games frequently used*, and 3 means *video games daily used*. The subjects who answered 0 were not included in the three groups to avoid that their lack of familiarity with video games could have a negative impact on the results of the experiment. The subjects were distributed among the three groups T, A, and B in the most possible uniform way, trying to balance age, school years, and ability in playing video games. This balance is important because the performance in the *PASAT* test mainly correlates with education and age, while the ability with video games was considered to avoid that a lower affinity of some subjects could influence the results obtained during the interaction with MS-rehab.

Group	*Age (avg/stdev)*	*Gender (#F/#M)*	*School years (avg/stdev)*	*VG ability (avg/stdev)*
T	23.7 / 1.89	5/5	16 / 1.73	1.7 / 0.67
A	24.3 / 2.87	6/4	16.2 / 1.83	1.9 / 0.74
B	23.8 / 1.99	5/5	16.2 / 1.83	1.8 / 0.63

Table 2. Demography of the three groups of subjects involved in the experiment.

The *Paced Auditory Serial Addition Test (PASAT)* [29] aims at measuring a subject's speed while processing controlled information, her/his working memory, and her/his sustained attention. The test is administered by proposing a sequence of 61 auditory numeric stimuli (numbers from 1 to 9) to the subject, who has to continuously calculate and report the sum of the last two heard numbers. For example: if the numbers 1 and 9 are presented, the correct sum to be reported is 10; if the next number is 4, the next correct answer will be 13, and so on until the end of the sequence.

The score for the *PASAT* is the total number of correct answers, the maximum score is 60. The version of *PASAT* used in our experiment [49] consists of audio files containing pre-recorded series of numbers in a random order. Each audio file corresponds to a different stimulus presentation frequency: the time interval between numbers can be 4000, 3000, 2600, 2200, or 1800 ms, while the pronunciation duration of each number is constant at about 600 ms. It is intuitive that the difficulty of the task increases as the time interval between numbers decreases. The test was administered according to the guidelines in [49], using a delay between stimuli equal to 2600 ms; this choice was made because the 4000 and 3000 ms versions were considered too simple for young healthy students and therefore not sufficient to highlight a possible cognitive improvement produced by the training software, while the 2200 ms and 1800 ms versions were deemed too difficult.

4.1.2 Experiment results. The students in groups A and B executed the *PASAT* test before their cognitive training. The average score of group A was 47.8, while that of group B was 49.2. We did not find a statistically significant difference between the two average scores (*2-tailed T-test*,[5] $p = 0.42$), meaning that the two groups belonged to the same population.

After groups A and B executed the cognitive training, we measured some indicators to highlight possible differences between the two groups, as summarized in Table 3. The average score of group A in the *PASAT* test was 55.5, therefore it increased by 7.7 points. That of group B was 52, therefore it increased by 2.8 points. Both groups performed significantly better than before intervention (*2-tailed paired T-test*, $p = 1.7 * 10^{-5}$ for group A, and $p = 0.02$ for group B). Moreover, subjects in group A performed significantly better than those in group B (*2-tailed T-test*, $p = 4 * 10^{-4}$). Subjects in group A failed, on average, the 0.56% of the exercises, while those in group B failed the 1.10% of the exercises. We did not find a statistically significant difference between the two groups (*2-tailed T-test*, $p = 0.56$). On average, group A made 0.85 error per exercises, while group B made 0.69 errors per exercise. Also in this case we did not find a statistically significant difference between the two groups (*2-tailed T-test*, $p = 0.56$). Group A omitted 0.11 responses per exercise, while group B omitted 0.18 responses per exercise. We did not find a statistically significant difference between the two groups also on this variable (*2-tailed T-test*, $p = 0.31$). It should be noted that performing multiple T-tests, as we did, could have resulted in an increased risk of Type 1 errors. However, even applying the Bonferroni correction to the p-values in the table, the statistical significance of the results does not change.

One week after the cognitive training, an alternating attention exercise and a working memory exercise at the maximum difficulty level were administered to both groups of students. As far as the alternating attention exercise is concerned, the difference between the two average performances was 0.003 points in favour of group B, while for the memory exercise it was 0.03 points in favour of

[5]We decided to use the T-test because it is generally considered robust against violations of the normality assumption [45].

Group	*PASAT score increase (avg/stdev)*	*Failed exercises % (avg/stdev)*	*Errors per exercise (avg/stdev)*	*Omissions per exercise (avg/stdev)*
A	7.7/2.95	0.56/0.07	0.85/1.93	0.11/0.38
B	2.8/3.26	1.10/0.11	0.69/1.71	0.18/0.58
2-tailed T-test (df=18)	$p = 4 * 10^{-4}$	$p = 0.56$	$p = 0.56$	$p = 0.31$

Table 3. Hypothesis 1: experiment results.

group A. By applying a *2-tailed T-test*, we did not find a statistically significant difference between the two groups for the alternating attention exercise ($p = 0.33$), nor for the working memory exercise ($p = 0.06$).

We then compared the baseline policy with the one learned by the RL system. First of all, we compared the policies for the attention exercise. In the baseline policy, as previously stated, after two consecutive passed exercises, the difficulty is increased. This is done by randomly selecting, with uniform probability, a parameter of the exercise, the current value of which is increased. Therefore, as there are 5 parameters, each parameter may be increased with a probability of 20% at any state of the training, except when it has reached its maximum value. Looking at the learned policy with RL, the situation is not much different, even though the probability distribution of the actions in each state is not perfectly uniform. As an illustration of the differences, Table 4 reports the probabilities assigned to the increase of each parameter by the baseline policy and the learned policy in the initial state of the attention exercise.

It should be noted that in Table 4 and all the following tables where we compare probabilities of different policies, we omitted the probabilities of all the actions aimed at decreasing a parameter. We made this decision for two reasons, the first being that these actions would be incomparable to the expert policy, which does not include actions that decrease parameters, and the second being that the probabilities of these actions are negligible (i.e., close to zero) due to the nature of our subjects (healthy students who almost never failed or performed poorly on an exercise). For the same reason, we also omitted the probability of the action that leaves all the parameters unmodified.

Parameter	*Expert policy*	*Category policy*
# Iteration	0.2	0.18
# Stimuli per target	0.2	0.14
Time	0.2	0.24
Distractors	0.2	0.21
Color	0.2	0.20

Table 4. Probabilities assigned by the expert policy and the learned category policy to increasing the parameters of the attention exercise in the initial state (all parameters set to their easiest value). Learned probabilities are rounded.

We then compared how many students of the two groups executed the attention exercise instances in the training, and the average performance they obtained, as shown in Figure 4. As long as all the students in A and B executed the exercises, their average performances were overlapping and remained close to 1. Then, after the execution of about 10 exercises, the students in A rapidly concluded their training, (none of them executed more than 15 exercises), and their average performance generally maintained a very high value. Differently, students belonging to group B executed more exercises, and their performance notably dropped when the number of exercises done reached the highest values. Regarding the total number of exercises, group A performed 119 exercises while group B performed 222 exercises (an average of 11.9 exercises per student for group A and 22.2 for group B)[6].

[6]It should be noted that the total number of exercises for group B includes the repetitions needed to meet the requirement to move to a more difficult exercise posed by the expert policy. That means that, in case of a failed exercise, a patient is forced to do at least three exercises (the failed one and then other two successful exercises in a row) to move to the next level.

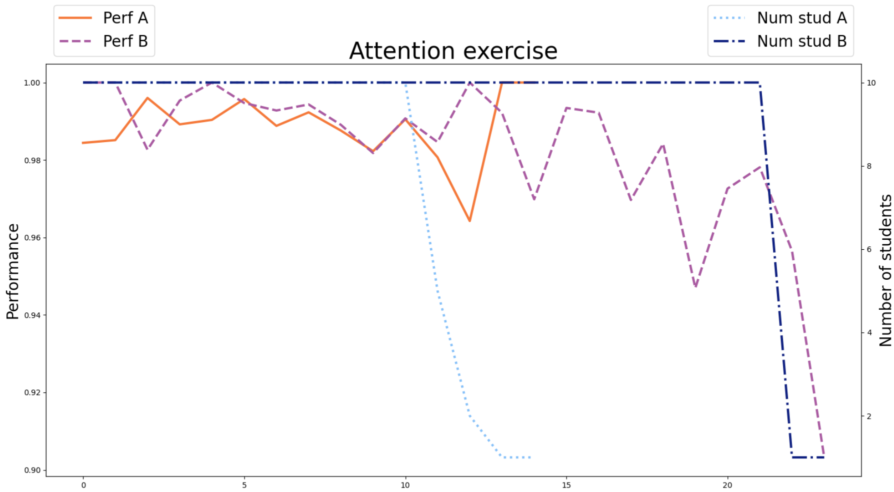

Fig. 4. Groups A and B along the training with the attention exercise. On the x axis, we have the number of executed exercise instances. On the right y axis, we have the number of students that performed a given number of exercises (light blue dotted line for group A, dark blue dash-dotted line for group B). On the left y axis, we have the average performance of these students (orange solid line for group A, purple dashed line for group B).

Moving on to the memory exercise, the situation is completely different. In this case the baseline policy is deterministic and the parameters are increased in this order: time, color and number of stimuli. When all these parameters reach their maximum value, the number of figures to be remembered is increased, and they are set back to the minimum value, before being re-incremented in the same order. Instead, the learned policy can upgrade any parameter in any state and not in a strictly defined fashion. Table 5 shows the great difference between the two policies. As the *PASAT* test demonstrated, this new, learned policy proved to be better than the experts' one. This is likely due to the fact that it allows a faster progression up the difficulty scale, allowing a user who does not need a lower level exercise to access more difficult exercises that actually fall within her/his training window.

Group A performed 243 memory exercises in total (24.3 exercise per subject on average), while group B performed 301 exercises (30.1 exercise per subject) on average[7]. When tracing a graph for the memory exercise (see Figure 5) analogous to that for the attention exercise, we note a similar, less evident, behavior. Also in this case the students in group A completed their training sessions globally before those in group B. Regarding the performance, all the students in A maintained an elevated value along the training, while the students in B who executed the highest number of exercises incurred into a general degradation of performance.

4.2 Hypothesis 2

After the first experiment, we conducted another test to assess Hypothesis 2.

4.2.1 Experiment design. We gathered a new group of ten students (called group C), with the same demographics of groups A and B. In particular, they were 6 females and 4 males, and their *Age,*

[7] Again, the total number of B includes the repetitions imposed from the experts policy rules and eventual failures.

Parameter	*Expert policy*	*Category policy*
# Figures	0.0	0.60
# Stimuli	0.0	0.11
Time	1.0	0.15
Color	0.0	0.12

Table 5. Probabilities assigned by the expert policy and the learned category policy to increasing the parameters of the memory exercise in the initial state (all parameters set to their easiest value). Learned probabilities are rounded.

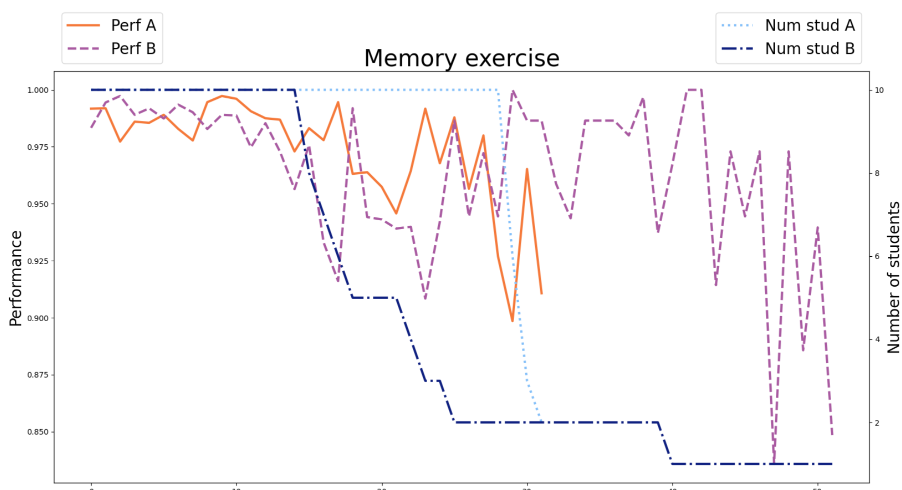

Fig. 5. Groups *A* and *B* along the training with the memory exercise. On the x axis, we have the number of executed exercise instances. On the right y axis, we have the number of students that performed a given number of exercises (light blue dotted line for group *A*, dark blue dash-dotted line for group *B*). On the left y axis, we have the average performance of these students (orange solid line for group *A*, purple dashed line for group *B*).

School years, and *VG ability (avg/stdev)* were 24.4/2.88, 16.8/1.40, and 2.0/0.82, respectively. We gave this group a suite of exercises identical to that of the previous groups. The only difference with group *A* was that the RL-agent had been reactivated, and thus each student in group *C* started from the student policy used by all the subjects in *A* but ended with a new, fully personalized, policy. As for the previous experiment, the subjects went through a run of the *PASAT* test before and after the training intervention, to measure the effects of the cognitive training. We measured also the other indicators mentioned in Section 4.1.

4.2.2 *Experiment results.* The average *PASAT* score of group *C* before the training was 49.5, while that measured after it was 56.8. As a preliminary evaluation, we compared the increase in the *PASAT* score of group *C* with that of group *B* (students who were trained with the baseline MS-rehab system), and discovered that the average cognitive functioning of group *C* after the treatment was significantly better than that of group *B*. This result gives and additional proof of the validity of Hypothesis 1.

Then, we moved on to the evaluation of Hypothesis 2, and compared the results of group *C* to those of group *A*, in order to see if the personalized policy actually led to better results than the category one. A summary of the results is shown in Table 6. These results show that the two groups are statistically equivalent on all the indicators apart from the average number of errors per exercises, where group *C* is significantly better (*2-tailed T-test,* $p = 1.23 * 10^{-5}$)[8]. This does not allow to conclude that a personalized policy is better than a category one to improve the cognitive functioning of students, but at least it induces less errors in the exercises. Moreover, it must be noted that group *A* already had an average score (55.5) that was very close to the highest achievable one (60), so it is extremely difficult for a personalized policy to do even better. This could be related to the fact that both groups *A* and *C* are formed by healthy and young people with high level of education.

The number of memory exercises required for group *C* to achieve an improvement substantially equivalent to that of group *A* was significantly lower (*2-tailed T-test,* $p = 4.87 * 10^{-5}$). In fact, the total number of exercises for group *C* was 46, while for group A it was 243, thus 4.6 and 24.3 on average for each individual respectively. On the other hand, no significant difference (*2-tailed t-test,* $p = 0.16$) was found in the number of attention exercises (132 total exercises for group *C* versus 119 for group *A*, thus an average of 13.2 and 11.9 respectively). Thus, we have a clue that the customised policy may, at least, be more efficient with regard to the number of exercises and, consequently, the time needed for training.

Looking at the new, personalized, learned policies, we did not find particular differences with respect to the category policies analyzed before. For both exercises, a summary of the probabilities of increasing the exercise parameters in some states is presented in Table 7 and Table 8. The personalized policies of the attention exercise are still pretty uniform for all the 10 students in group *C*, while in the memory exercise we still have a high probability for the action that increases the number of images. As can be seen in Table 8, the personalised policies of group *C* maintain a higher probability of increasing the number of images to be remembered in the memory exercise also in the second state; this led to the great advantage of group *C* over group *A* with regard to the number of exercises needed to complete the session, which is the greatest advantage we have seen in the use of a personalized policy and which may be the reason for the reduced number of failed exercises and errors per exercise.

Analogously to what done before, we compared how many students of groups *A* and *C* executed the attention exercise instances in the training, and the average performance they obtained. The

[8]Even if we apply the Bonferroni correction to p-values in the table, the statistical significance of the results does not change.

Group	*PASAT score increase (avg/stdev)*	*Failed exercises % (avg/stdev)*	*Errors per exercise (avg/stdev)*	*Omissions per exercise (avg/stdev)*
A	7.7/2.95	0.56/0.07	0.85/1.93	0.11/0.38
C	7.3/1.70	0.0/0.0	0.26/0.53	0.12/0.37
2-tailed T-test (df=18)	$p = 0.84$	$p = 0.30$	$p = 1.23 * 10^{-5}$	$p = 0.98$

Table 6. Hypothesis 2: experiment results.

Parameter	*# Iteration*	*# Stimuli per target*	*Time*	*Distractors*	*Color*
Category policy	0.18	0.14	0.24	0.21	0.20
Subject 1	0.14	0.18	0.22	0.21	0.23
Subject 2	0.23	0.20	0.16	0.15	0.23
Subject 3	0.27	0.23	0.23	0.09	0.17
Subject 4	0.13	0.25	0.15	0.20	0.25
Subject 5	0.21	0.15	0.21	0.24	0.16
Subject 6	0.25	0.14	0.21	0.22	0.18
Subject 7	0.18	0.23	0.14	0.22	0.19
Subject 8	0.24	0.16	0.25	0.10	0.23
Subject 9	0.16	0.21	0.22	0.20	0.19
Subject 10	0.12	0.12	0.28	0.26	0.20

Table 7. Probabilities assigned by the category policy and the personalized learned policies for each student in group C to increasing the parameters of the attention exercise in the initial state (all parameters set to their easiest value). All probabilities are rounded.

comparison is shown in Figure 6. We can see that, for the attention exercise, the performances overlap up to the 15^{th} completed exercise, then group C had a drop in performance around the 18^{th} exercise, which is then recovered in the next one. It should be noted that only one subject reached this high number of exercises, so this strong fluctuation may have occurred by chance.

Turning to the memory exercise (see Figure 7), group C experienced a drop in performance around the 8^{th} exercise, immediately followed by near-optimal performance. Group A, instead, had a higher performance at the beginning, which then decreased slightly over time and returned to higher performance around the 30^{th} exercise. It should also be noted that after the 45^{th} exercise, there was another drop in performance, but again this only affected one user.

4.3 Discussion

The results of our first experiment (Section 4.1) support the validity of Hypothesis 1. The improvement in the neuropsychological *PASAT* test was significant for both groups A and B of students executing the cognitive training with the MS-rehab system. In addition, group A (trained using a version of MS-rehab embedding a difficulty-variation policy learned with RL from other students having the same characteristics) performed significantly better than group B (trained with a MS-rehab based on a difficulty-variation mechanism defined by experts in cognitive rehabilitation). This demonstrates the effectiveness of the exercise category policies learned using RL for the two exercises selected for the experiment, and their superiority with respect to a state-of-the-art difficulty-variation procedure. The policy defined by the expert was shown to be suboptimal in the experiment, because it does not adapt to the individual the way the exercise parameters are changed. Moreover, since a subject needs to pass two exercises in a row in order to get an increase of the difficulty, it does not take into account the difficulty of the last completed exercise. Indeed, if an exercise is trivial for a subject, it could be skipped as soon as it is passed, while if it is tricky, the difficulty should be increased more gradually. On the contrary, an adaptive policy can solve this problem considering the absolute performance of the last exercise, which could be perfect (100%), barely sufficient (80%), insufficient (60-80%), or badly insufficient (<60%).

The goal of the second experiment we performed (Section 4.2) was to show that a personalized policy can outperform the category policy from which it is refined by the second phase of our

State 1

Parameter	*# Figures*	*# Stimuli*	*Time*	*Color*
Category policy	0.60	0.11	0.15	0.12
Subject 1	0.65	0.05	0.10	0.19
Subject 2	0.64	0.10	0.10	0.14
Subject 3	0.88	0.05	0.05	0.01
Subject 4	0.65	0.14	0.15	0.05
Subject 5	0.67	0.16	0.12	0.03
Subject 6	0.76	0.10	0.09	0.03
Subject 7	0.80	0.06	0.03	0.10
Subject 8	0.66	0.07	0.16	0.10
Subject 9	0.60	0.15	0.17	0.07
Subject 10	0.72	0.09	0.10	0.07

State 2

Parameter	*# Figures*	*# Stimuli*	*Time*	*Color*
Category policy	0.33	0.31	0.20	0.14
Subject 1	0.52	0.03	0.20	0.22
Subject 2	0.43	0.19	0.13	0.22
Subject 3	0.55	0.20	0.10	0.12
Subject 4	0.60	0.15	0.16	0.07
Subject 5	0.53	0.10	0.20	0.16
Subject 6	0.50	0.26	0.07	0.15
Subject 7	0.49	0.14	0.10	0.26
Subject 8	0.62	0.17	0.11	0.08
Subject 9	0.51	0.08	0.03	0.36
Subject 10	0.56	0.17	0.06	0.19

Table 8. Probabilities assigned by the category policy and the personalized learned policies for each student in group C to increasing the parameters of the attention exercise in the initial state (all parameters set to their easiest value) and in the second state (all parameters set to their easiest value but # figures, which has already been incremented). All probabilities are rounded.

policy-building method (Hypothesis 2). We did not gather evidence of that from the point of view of the cognitive functioning measured by the *PASAT* test. This result could be caused by the subjects involved in our experiments (i.e., young educated students), who reached a cognitive performance that was very close to the highest achievable also without a fully personalized policy. In order to have a better understanding of this aspect of our policy-building method, new experiments must be performed involving different populations (e.g., elders, or cognitive impaired people, or teenagers with special needs). However, the second experiment showed a net advantage in terms of efficiency of the training process, as the training with the personalized policies involved the execution of a significantly inferior number of memory exercises compared to the category policy for achieving a comparable *PASAT* score.

We noticed a very different pattern in the increase of difficulty in the two exercises. In the attention exercise, the parameters were increased rather uniformly, whereas in the memory exercise much more emphasis was placed on the number of images to be remembered, which was increased much

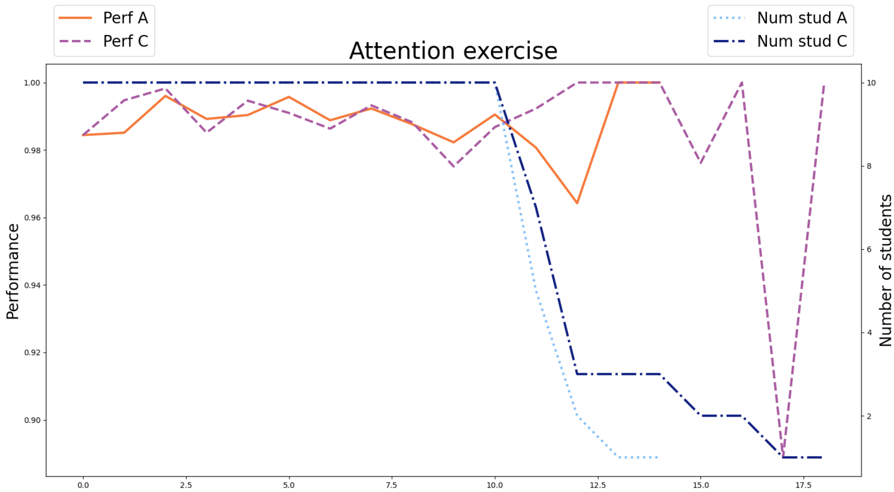

Fig. 6. Groups A and C along the training with the attention exercise. On the x axis, we have the number of executed exercise instances. On the right y axis, we have the number of students that performed a given number of exercises (light blue dotted line for group A, dark blue dash-dotted line for group C). On the left y axis, we have the average performance of these students (orange solid line for group A, purple dashed line for group C).

more often than the other parameters. This was reflected in the number of completed exercises until the end of the cognitive training. The average number of memory exercises for group A was 24.3, for group B it was 30.1, and for group C it was 5. This means that the RL-agent was able to find that this parameter is critical to adapt the memory exercise so that cognitive training is faster, with equal effectiveness.

The results of our experiments show that a (category or personalized) difficulty variation policy learned with RL can outperform an expert-designed policy especially when the latter is highly deterministic in the way it changes the exercise parameters (as in the case of the memory exercise), while it can be useful to adjust it when it is probabilistic (as in the case of the attention exercise).

Finally, it is important to note that the results of the *PASAT* test may have been influenced by a *practice effect* [52]. This means that a subject taking the test for the second time after cognitive training might obtain a better result not because her/his cognitive functioning has improved, but for other reasons, such as reduced anxiety on the repeated test, and/or a better understanding of the task. However, we tend to exclude that the practice effect influenced the results of our experiments because there is evidence that practice does not affect the specific subjects involved, i.e., university students [49].

5 CONCLUSIONS AND FUTURE WORK

Adapting the difficulty of exercises to the level of treated subjects is a fundamental requirement for the effectiveness of cognitive training or rehabilitation. Authors of state-of-the-art software systems dedicated to this purpose claim that their tools provide automatic adaptive mechanisms to fit the difficulty of exercises to how the subjects perform in their execution. However, we have highlighted that the degree of adaptability of these systems is still limited to predefined rules for

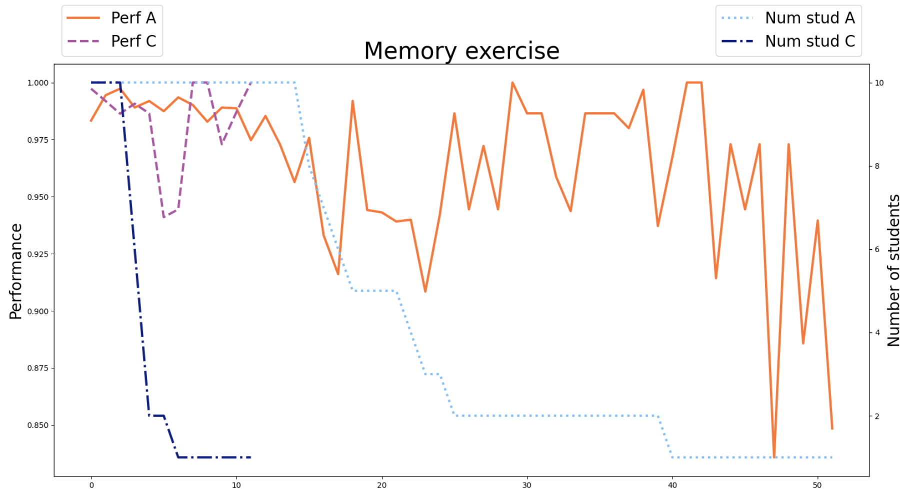

Fig. 7. Groups A and C along the training with the memory exercise. On the x axis, we have the number of executed exercise instances. On the right y axis, we have the number of students that performed a given number of exercises (light blue dotted line for group A, dark blue dash-dotted line for group C). On the left y axis, we have the average performance of these students (orange solid line for group A, purple dashed line for group C).

the progression of difficulty levels that automatically change the parameters influencing difficulty according to a fixed strategy, identical for all subjects.

We have designed an adaptive mechanism based on Reinforcement Learning that overcomes these limitations. The policy used by the system to automatically change the difficulty level of the exercises is learned while the individuals execute the training, by exploiting their performance as a reward indicating the goodness of the way the difficulty was previously modified. We have embedded the proposed RL mechanism into MS-rehab, a computerized cognitive training system that provides a set of exercises suitable for the treatment of mild cognitive dysfunctions mainly caused by multiple sclerosis, but also usable for the treatment of other diseases and for the cognitive training of healthy subjects.

The results of the two experiments we conducted allow us to draw interesting conclusions. As a first achievement, we have shown that it is possible to learn, using RL, an effective category policy to be used in the cognitive training process of university students. This policy led to a significant improvement in the neuropsychological *PASAT* test, which was substantially more relevant than the improvement obtained using a policy to guide the training process designed by cognitive rehabilitation experts.

In our second experiment, we showed that the category policy learned for students can still be improved by allowing the RL algorithm to further learn a customised policy for an individual using her/his results during the training process. Although we were not able to further improve the results obtained in the *PASAT* test by the involved university students, we were able to help them achieve equivalent results in a much shorter time and with far fewer exercises.

The solution we propose is a significant advance on the state of the art, in that an individual can effectively practice cognitive training at a personalised level after an initial adaptation phase even without the intervention of a therapist. It would be particularly suitable to support collective training sessions, which are a common practice in many situations, e.g. rehabilitation in mental illness or chronic degenerative diseases. Indeed, in collective sessions, it is almost impossible for therapists to monitor all subjects performing the exercises and thus correctly adjust the difficulty levels of their training.

Undoubtedly, the two-step method we propose for constructing adaptive cognitive training policies has proven its effectiveness when applied to university students, and will possibly work equally well for other categories of users. The future development of this research should investigate this in depth. In particular, we plan to strengthen the results about Hypothesis 1 by replicating the experiment we have done with students for other more critical categories of subjects (e.g., elderly people, or patient suffering of a mental illness or neuro-degenerative diseases). In addition, we want to investigate the influence of the degree of specificity of a category policy; for example, it could be interesting to verify if a policy learned for people suffering from a neuro-degenerative disease is still effective for the training of subjects suffering from another disease producing similar deficits, or if the category policy should be totally specific for every different pathology or syndrome.

We also need to set up experiments to verify Hypothesis 3, for which a policy elicited for a category would not improve the training process of subjects belonging to another category, if applied for their cognitive training. The same has to be done to verify Hypothesis 4, for which an individually personalized policy for a subject belonging to a category can be more efficiently learned refining a policy previously learned for her/his category instead of starting from a fully random policy.

Another very interesting direction of research is investigating other ways to calculate the reward given to the RL-agents learning the adaptive difficulty-variation policies. In this work, we have adopted the simple model for calculation described in Section 3.2. Other models are possible, for example the *Rash model* [19], which has been used in education and healthcare to measure the performance of subjects in psychometric tests.

On a technical standpoint, it could be very useful to exploit techniques like *teacher advice* and *transfer learning* [51] in order to ease the learning of new difficulty-variation policies from those already elicited using our approach.

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